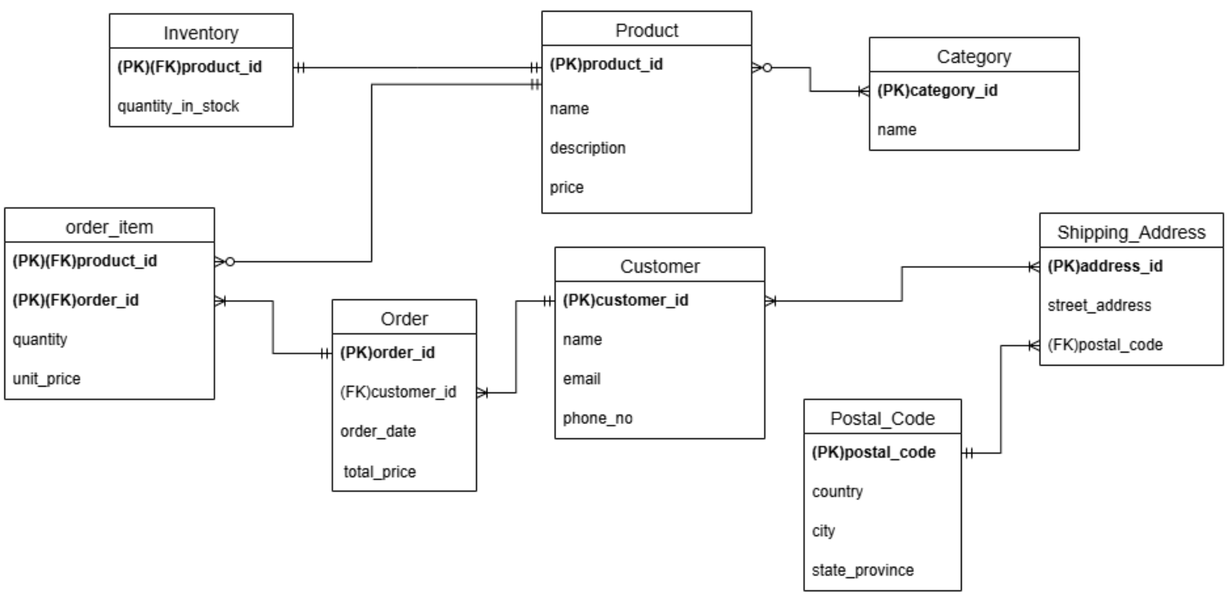
Assignment DB - Relational Database Design and Optimization

1. Assumptions

- Product to Category Relationship(M: N) - A Product can be categorized into one or many categories. A category can have zero or many products. (eg, TV categories – Electronics, Home Appliances)
- Order Item Entity – Unit price means the price of that item when it is ordered. Using the price attribute of the product entity for calculations can give incorrect outputs. Because the price attribute of the Product entity can be changed in future.
- Customer to Shipping Address Relationship(M: N) – Customer can have different shipping addresses. And there can be situations where each member of the same house(same address) uses the same address.
- Postal Code Entity – Country, city, and state_province attributes are dependent on the postal code. That is why it has a separate entity.

2. Logical ERD



3. Relational Schema

- Join tables are included in between M: N relationships(Product to Category, Customer to Shipping Address).
- BINARY(16) UUID is used for primary keys for entities that can grow hugely in the future.
- Use the int data type for the Category entity primary key since its categories do not grow as much as compared to others.